DAY 1 · AWS LINUX LAB

AWS, SSH & Firewall

Foundations & hardening

Spun up a free-tier Ubuntu EC2 instance in eu-north-1, locked SSH down to key-only access from a single IP, and enabled UFW as a basic firewall.

This is the foundation the rest of the lab connects to — every later day assumes this instance, this key, and this hardened SSH setup are already in place.

Local — your machine | Remote — the EC2 box | Config file | Gotcha | Best practice | Paths

What we covered

- ✓ Launched Ubuntu Server 24.04 LTS, x86_64, `t2.micro` / `t3.micro` (free tier)
- ✓ Security group restricting SSH (port 22) to **My IP** only — not `0.0.0.0/0`
- ✓ Generated an **ED25519** key pair — faster, smaller and more modern than RSA
- ✓ Moved the key to `~/.ssh/` and renamed it to match the naming convention
- ✓ Set up a local `~/.ssh/config` so `ssh lab-ubuntu-01` connects with zero extra flags
- ✓ Hardened `sshd_config` : disabled root login and password auth
- ✓ Installed and enabled UFW, allowing only OpenSSH

Naming note · Day 1 resources use the old scheme (`aws-lab-ubuntu-01`). Under the current rule `<type>-<workload>-<env>-<region>-<instance>` the key would be `kp-ubuntu-lab-eun1-001` and the instance `vm-ubuntu-lab-eun1-001` .

Full command sequence

Local — Key file setup — move into ~/.ssh and rename

```
mkdir -p ~/Projects/aws-lab                              # -p: create folder, no error if it exists
mv ~/Documents/aws/ubuntu-lab-key.pem ~/.ssh/            # move key into ~/.ssh
chmod 400 ~/.ssh/ubuntu-lab-key.pem                      # 400: owner read-only (required by SSH)
mv ~/.ssh/ubuntu-lab-key.pem ~/.ssh/aws-lab-ubuntu-01.pem   # rename to naming convention
```

Local — First connection — before the alias existed

```
ssh -i ~/.ssh/aws-lab-ubuntu-01.pem ubuntu@<instance-ip>      # -i: which private key to use
```

Local ~/.ssh/config — connect without flags

```
nano ~/.ssh/config                                  # open (or create) the client config
```

~/.ssh/config

```
Host lab-ubuntu-01                                  # alias you type after ssh
    HostName <instance-ip>                          # real address (changes on stop/start)
    User adm-fkar                                   # login user, adm-<name> rule
    IdentityFile ~/.ssh/aws-lab-ubuntu-01.pem       # key used automatically
```

Gotcha · SSH config files don't support comments at the end of a line — a `#` only works at the start of its own line. That's why the notes sit beside the config here instead of inside it.

```
chmod 600 ~/.ssh/config                             # 600: owner read/write only
ssh lab-ubuntu-01                                   # connect using the alias
```

Remote Fix terminal rendering (Kitty terminfo)

```
export TERM=xterm-256color                          # fixes "Error opening terminal: xterm-kitty"
```

Remote Harden sshd_config

```
sudo nano /etc/ssh/sshd_config                      # edit the SSH server config as root
```

/etc/ssh/sshd_config

```
PermitRootLogin no                                  # nobody logs in directly as root
PasswordAuthentication no                           # keys only, no password guessing
PubkeyAuthentication yes                            # allow key-based login
```

```
sudo sshd -t                                        # -t: test config syntax BEFORE restarting
sudo systemctl restart ssh                          # apply the changes
```

Remote UFW firewall

```
sudo ufw allow OpenSSH                              # allow SSH before enabling, avoids lockout
sudo ufw enable                                     # turn the firewall on
sudo ufw status                                     # confirm the active rules
```

Remote Close the session

```
exit                                                # or Ctrl+D
```

Where things live

What	Path
Private key (local)	`~/.ssh/aws-lab-ubuntu-01.pem`
Local SSH config	`~/.ssh/config`
SSH daemon config (remote)	`/etc/ssh/sshd_config`
SSH config overrides (remote)	`/etc/ssh/sshd_config.d/`
Original key download (moved out of)	`~/Documents/aws/`

Best practices

Scope SSH to a known IP
Never leave port 22 open to `0.0.0.0/0`.

ED25519 over RSA
Smaller, faster, and no known weaknesses for new keys.

Uncommenting ≠ changing
Removing `#` keeps the old value — edit the value itself.

Test before restart
Run `sshd -t` before restarting anything that could lock you out.

Keep a fallback session
Leave one SSH session open while testing a config change.

Public IPs move
A stopped/started instance gets a new IP unless an Elastic IP is attached.

Keys live in ~/.ssh/
Not in a project folder — that's where SSH tooling expects them.

Lock down permissions
`400` on keys, `600` on config — SSH rejects looser ones.

Part of the AWS/Linux sysadmin lab · see day1-naming-conventions for the naming reference